

MCQs: Encoding Schemes and Introduction (Class XI)

1. What does a computer fundamentally understand?

- A. Decimal numbers
- B. Alphabetic characters
- C. Binary language of 0s and 1s
- D. Human-readable symbols

Answer: C

2. When a key is pressed on the keyboard, what happens first inside the computer?

- A. It is directly converted to binary
- B. It is mapped to a unique code value
- C. It is stored as text
- D. It is displayed on the monitor

Answer: B

3. The unique code assigned to a key is later converted into:

- A. Decimal value
- B. Octal value
- C. Binary value
- D. Hexadecimal value only

Answer: C

4. When the key 'A' is pressed, it is internally mapped to which decimal value?

- A. 64
- B. 65
- C. 66
- D. 97

Answer: B

5. The decimal value 65 is converted into binary so that:

- A. Humans can understand it
- B. Printers can print it
- C. The computer can understand it
- D. It can be encrypted

Answer: C

6. When the Hindi letter ‘अ’ is pressed, it is internally mapped to which hexadecimal value?

- A. 0041
- B. 0065
- C. 0905
- D. 0915

Answer: C

7. What is the binary equivalent of hexadecimal value 0905 as given in the text?

- A. 0000010100001001
- B. 0000100100000101
- C. 1001000001010000
- D. 0000100101010000

Answer: B

8. What is meant by “encoding”?

- A. Storing data permanently
- B. Encrypting data for secrecy
- C. Converting data into an equivalent cipher using a specific code
- D. Compressing data

Answer: C

9. Why is the same code value used for a key across all keyboards?

- A. Due to hardware similarity
- B. Due to software dependency
- C. Due to standard encoding schemes
- D. Due to operating system control

Answer: C

10. Standard encoding schemes ensure that:

- A. Computers work faster
- B. Characters have unique code values
- C. Memory usage is reduced
- D. Data remains encrypted

Answer: B

11. ASCII stands for:

- A. American Scientific Code for Information Interchange
- B. American Standard Code for Information Interchange
- C. Advanced Standard Code for Internet Interface
- D. Automated System Code for Information Interchange

Answer: B

12. Why was ASCII developed in the early 1960s?

- A. To increase memory capacity
- B. To reduce hardware cost
- C. To standardise character representation
- D. To support Indian languages

Answer: C

13. Before ASCII, computers had difficulty communicating because:

- A. They used different operating systems
- B. They used different keyboards
- C. They represented keyboard characters differently
- D. They lacked binary representation

Answer: C

14. ASCII is still widely used because:

- A. It supports all world languages
- B. It is simple and standardised
- C. It uses 16 bits
- D. It replaces Unicode

Answer: B

15. Initially, ASCII used how many bits to represent characters?

- A. 6 bits
- B. 7 bits
- C. 8 bits
- D. 16 bits

Answer: B

16. How many characters can be represented using 7-bit ASCII?

- A. 64
- B. 100
- C. 128
- D. 256

Answer: C

17. ASCII is capable of encoding characters of which language only?

- A. All languages
- B. Indian languages
- C. English language
- D. European languages

Answer: C

18. What is the ASCII decimal value of capital letter 'A'?

- A. 64
- B. 65
- C. 97
- D. 66

Answer: B

19. What is the ASCII decimal value of lowercase letter 'a'?

- A. 65
- B. 66
- C. 96
- D. 97

Answer: D

20. What is the ASCII decimal value of the space character?

- A. 0
- B. 16
- C. 32
- D. 48

Answer: C

21. The ASCII value of 'D' is:

- A. 66
- B. 67

- C. 68
- D. 69

Answer: C

22. What is the 7-bit binary equivalent of ASCII value 68?

- A. 1000001
- B. 1000100
- C. 1010100
- D. 1001000

Answer: B

23. Which word is encoded as an example in the text?

- A. CODE
- B. DATA
- C. ASCII
- D. BYTE

Answer: B

24. What is the ASCII value of 'T' in the word DATA?

- A. 82
- B. 83
- C. 84
- D. 85

Answer: C

25. What is the binary equivalent of ASCII value 84?

- A. 1000100
- B. 1000001
- C. 1010100
- D. 1100100

Answer: C

26. Cipher refers to:

- A. Plain text
- B. Human-readable form
- C. Coded form of data
- D. Binary data only

Answer: C

27. Encryption means:

- A. Storing data
- B. Converting data to cipher
- C. Printing data
- D. Displaying data

Answer: B

28. Decryption means:

- A. Destroying data
- B. Compressing data
- C. Converting cipher back to original content
- D. Encoding data

Answer: C

29. ISCII was developed to support:

- A. European languages
- B. Programming languages
- C. Indian languages
- D. Machine languages

Answer: C

30. ISCII was developed during:

- A. Early 1960s
- B. Late 1970s
- C. Mid 1980s
- D. Early 2000s

Answer: C

31. ISCII is an:

- A. 7-bit code
- B. 8-bit code
- C. 16-bit code
- D. 32-bit code

Answer: B

32. How many characters can an 8-bit code represent?

- A. 128
- B. 200
- C. 256
- D. 512

Answer: C

33. ISCII retains how many ASCII codes?

- A. 64
- B. 96
- C. 128
- D. 256

Answer: C

34. Additional Indian language characters in ISCII are placed in which range?

- A. 0–127
- B. 64–191
- C. 128–159
- D. 160–255

Answer: D

35. The term 'aksharas' refers to:

- A. Binary digits
- B. English alphabets
- C. Indian language characters
- D. ASCII symbols

Answer: C

36. Why was UNICODE developed?

- A. To replace ASCII only
- B. To speed up processing
- C. To incorporate all written languages
- D. To reduce memory usage

Answer: C

37. Different encoding schemes failed earlier because:

- A. They used same codes
- B. They were hardware dependent
- C. They could not communicate with each other
- D. They supported Unicode

Answer: C

38. UNICODE assigns:

- A. Same number to multiple characters
- B. Unique number to every character
- C. Binary numbers only
- D. Decimal values only

Answer: B

39. UNICODE is independent of:

- A. Device
- B. Operating system
- C. Software application
- D. All of the above

Answer: D

40. Which of the following are common UNICODE encodings?

- A. UTF-7, UTF-9
- B. UTF-8, UTF-16, UTF-32
- C. ASCII-8, ASCII-16
- D. ISCII-16, ISCII-32

Answer: B

41. UNICODE is a superset of:

- A. ISCII
- B. Binary
- C. ASCII
- D. Hexadecimal

Answer: C

42. In UNICODE, values from 0–128 are:

- A. Reserved
- B. Empty
- C. Same as ASCII
- D. Same as ISCII

Answer: C

43. Unicode values for Devanagari script are represented in:

- A. Decimal
- B. Binary
- C. Hexadecimal
- D. Octal

Answer: C

44. Each Unicode table cell contains:

- A. Only characters
- B. Only code values
- C. Character with its hexadecimal value
- D. Binary representation

Answer: C

45. Why does UTF-32 take more space than UTF-16 or UTF-8?

- A. It uses fixed 32 bits for each character
- B. It supports fewer characters
- C. It uses ASCII only
- D. It compresses data

Answer: A

46. Integrated Circuits (ICs) in a computer are made up of a large number of:

- A. Capacitors
- B. Diodes
- C. Transistors
- D. Resistors

Answer: C

47. The ON (high) state of a transistor is represented by which digit?

- A. 0
- B. 1

- C. 2
- D. 9

Answer: B

48. The OFF (low) state of a transistor is represented by:

- A. 1
- B. 2
- C. 8
- D. 0

Answer: D

49. Which digits form the binary number system?

- A. 0, 1, 2
- B. 1, 2
- C. 0 and 1
- D. 0 to 9

Answer: C

50. Binary number system is also known as:

- A. Base-8 system
- B. Base-10 system
- C. Base-16 system
- D. Base-2 system

Answer: D

51. Which of the following is a valid binary number?

- A. 102101
- B. 111111.01
- C. 128.45
- D. 19A

Answer: B

52. Binary numbers can be converted into decimal numbers mainly because:

- A. Computers understand decimal
- B. Humans understand decimal better
- C. Decimal is faster
- D. Decimal needs less memory

Answer: B

53. Why was the octal number system introduced?

- A. To replace decimal system
- B. To reduce hardware cost
- C. For compact representation of binary numbers
- D. To encrypt data

Answer: C

54. Octal number system is known as:

- A. Base-2
- B. Base-4
- C. Base-8
- D. Base-16

Answer: C

55. How many digits are used in octal number system?

- A. 2
- B. 8
- C. 10
- D. 16

Answer: B

56. Which digits are allowed in octal number system?

- A. 0–9
- B. 1–8
- C. 0–7
- D. 0–15

Answer: C

57. How many binary digits are required to represent one octal digit?

- A. 2
- B. 3
- C. 4
- D. 8

Answer: B

58. The relationship $8 = 2^3$ explains:

- A. Why binary uses 3 bits
- B. Why octal uses 3 bits
- C. Why hexadecimal uses 3 bits
- D. Why decimal uses 3 bits

Answer: B

59. Base value of a number system is written as:

- A. Superscript
- B. Prefix
- C. Subscript
- D. Suffix

Answer: C

60. Which representation correctly shows an octal number?

- A. $(70)_{10}$
- B. $(70)_8$
- C. $(70)_2$
- D. $(70)_{16}$

Answer: B

61. Hexadecimal number system uses how many symbols?

- A. 8
- B. 10
- C. 12
- D. 16

Answer: D

62. Hexadecimal number system is called:

- A. Base-8
- B. Base-10
- C. Base-16
- D. Base-32

Answer: C

63. Which symbols are used in hexadecimal number system?

- A. 0–9 only
- B. A–F only
- C. 0–7 and A–F
- D. 0–9 and A–F

Answer: D

64. Decimal numbers 10 to 15 are represented in hexadecimal as:

- A. B to G
- B. A to F
- C. F to K
- D. 1A to 1F

Answer: B

65. How many binary bits are needed to represent one hexadecimal digit?

- A. 2
- B. 3
- C. 4
- D. 8

Answer: C

66. The relation $16 = 2^4$ explains:

- A. Why hexadecimal uses 4 bits
- B. Why octal uses 4 bits
- C. Why decimal uses 4 bits
- D. Why binary uses 4 bits

Answer: A

67. Hexadecimal symbol 'A' represents which decimal value?

- A. 9
- B. 10
- C. 11
- D. 12

Answer: B

68. What is the 4-bit binary equivalent of hexadecimal 'F'?

- A. 1010
- B. 1100
- C. 1110
- D. 1111

Answer: D

69. Why are hexadecimal numbers used for memory addresses?

- A. They are encrypted
- B. They are shorter and easier to remember
- C. They consume less power
- D. They are faster

Answer: B

70. A 16-bit binary memory address is difficult to manage because:

- A. It is too slow
- B. It is too large to display
- C. It is hard to remember
- D. It is hardware dependent

Answer: C

71. Binary address 1100000011110001 is written in hexadecimal as:

- A. C1F0
- B. C0F1
- C. CF01
- D. CFF1

Answer: B

72. Why are hexadecimal numbers used for webpage colour representation?

- A. Colours are binary
- B. Hexadecimal is faster
- C. Binary colour codes are too long
- D. Decimal cannot represent colours

Answer: C

73. RGB stands for:

- A. Red Green Blue
- B. Random Graph Blocks

- C. Red Grey Black
- D. Real Graphic Base

Answer: A

74. How many bits are required to represent one RGB colour?

- A. 8
- B. 16
- C. 24
- D. 32

Answer: C

75. Each RGB component (Red, Green, Blue) uses:

- A. 4 bits
- B. 6 bits
- C. 8 bits
- D. 16 bits

Answer: C

76. The hexadecimal code for RED colour is:

- A. (00,FF,00)
- B. (FF,00,00)
- C. (00,00,FF)
- D. (FF,FF,00)

Answer: B

77. Which colour has hexadecimal value (FF,FF,FF)?

- A. Black
- B. Grey
- C. Yellow
- D. White

Answer: D

78. Decimal number system is most commonly used by:

- A. Computers
- B. Digital systems
- C. Humans
- D. Microprocessors

Answer: C

79. To convert a decimal number to another number system, we repeatedly:

- A. Multiply by base
- B. Add base
- C. Divide by base
- D. Subtract base

Answer: C

80. In decimal to binary conversion, the base value used is:

- A. 10
- B. 8
- C. 16
- D. 2

Answer: D

81. While converting decimal to any other number system, remainders are written:

- A. Top to bottom
- B. Bottom to top
- C. Left to right
- D. Right to left

Answer: B

82. Binary equivalent of $(65)_{10}$ is:

- A. $(1001001)_2$
- B. $(1000100)_2$
- C. $(1000001)_2$
- D. $(1010001)_2$

Answer: C

83. To convert decimal to octal, the number is repeatedly divided by:

- A. 2
- B. 4
- C. 8
- D. 16

Answer: C

84. Octal equivalent of (65)₁₀ is:

- A. (101)₈
- B. (75)₈
- C. (145)₈
- D. (111)₈

Answer: A

85. Hexadecimal equivalent of (65)₁₀ is:

- A. (41)₁₆
- B. (14)₁₆
- C. (4A)₁₆
- D. (65)₁₆

Answer: B

86. Binary number is converted to octal by grouping bits in sets of:

- A. 2
- B. 3
- C. 4
- D. 8

Answer: B

87. Binary number is converted to hexadecimal by grouping bits in sets of:

- A. 2
- B. 3
- C. 4
- D. 8

Answer: C

88. If binary digits are not in perfect groups, what should be done?

- A. Ignore extra bits
- B. Add 1s
- C. Add 0s at MSB/LSB
- D. Stop conversion

Answer: C

89. While converting fractional binary to octal/hexadecimal, grouping is done:

- A. Right to left
- B. Left to right
- C. Randomly
- D. Top to bottom

Answer: B

90. To convert fractional decimal number to another system, we repeatedly:

- A. Divide by base
- B. Add base
- C. Multiply by base
- D. Subtract base

Answer: C

91. If fractional part starts repeating during conversion, we should:

- A. Continue forever
- B. Stop conversion
- C. Add zeros
- D. Ignore fraction

Answer: B

92. Binary equivalent of $(0.25)_{10}$ is:

- A. $(0.1)_2$
- B. $(0.01)_2$
- C. $(0.001)_2$
- D. $(0.11)_2$

Answer: B

93. Which number system is directly understood by the computer?

- A. Decimal
- B. Octal
- C. Hexadecimal
- D. Binary

Answer: D

94. Octal and hexadecimal systems are mainly used to:

- A. Increase storage
- B. Simplify binary representation
- C. Encrypt data
- D. Reduce processing time

Answer: B

95. Encoding schemes ultimately convert text into:

- A. Decimal form
- B. ASCII only
- C. Binary form
- D. Unicode form

Answer: C